



Birla Institute of Technology & Science, Pilani

K K Birla Goa Campus

Academic - Undergraduate Studies Division

FIRST SEMESTER, 2026–2027

Course Handout Part II

Course No.	CS F407
Course Title	Artificial Intelligence
Credits	3
Lecturers	Tirtharaj Dash (Room CC-107) and Ashwin Srinivasan (Room D-167)
Office Hours	By appointment (send an email to fix a time)

1. Lecture and Tutorial Details

Day	Time	Location	Activity
Tuesday	11:00–12:00	LT-2	Lecture
Wednesday	08:00–09:00	LT-2	Lecture/Tutorial
Friday	11:00–12:00	LT-2	Lecture
		Take-home	Project-related work, etc.

2. Course Description and Objectives

Pre-requisite (preferable): Data Structures and Algorithms.

On successful completion of the course, students should be able to:

1. understand the basic concepts underlying AI tasks, including problem solving, search, representation, reasoning, decision making, and learning;
2. understand and use modern AI tools (including LLMs) to address classical AI problems.

3. Course Plan

The number in parentheses denotes the approximate number of lectures for that topic.

Week	AI Science	AI Engineering	Labs: NewFAI meets GOF AI
1	Introduction (1)	Neural models (2)	Using LLMs to generate code for brute-force search, heuristic search, and constraint satisfaction
2	Agents and Search (2)	LLMs (1)	
3	Logical Intelligence (2)	Transformers (1)	
4	Planning (1)	LLM Agents (2)	QA with LLM Agents
5	Probabilistic Intelligence (1)	Extending LLMs: Tools (2)	
6	Probabilistic Intelligence (2)	Extending LLMs: RAG (1)	
7	Probabilistic Intelligence (2)	Fine-tuning (1)	
8	Neural Intelligence (2)	Beyond Transformers (1)	
9	Reward-based Intelligence (2)	RLHF (1)	Project

Week	AI Science	AI Engineering	Labs: NewFAI meets GOFAI
10	Learning (2)	Deep Learning (1)	
11	Guest lectures on Applications of AI (3)		
12	Project Presentations		
13	Project Presentations		

4. Evaluation Scheme

Assessment Component	Marks	Clarification
Worksheets/Labwork (best k -of- N)	20	This component consists of quizzes conducted during the semester. There will be N quizzes in total; the best k scores will count.
Comprehensive Exam	30	End-of-semester comprehensive examination.
Project	30	Divided between a mid-semester progress flash-talk and a final poster presentation.
Paper Reading	10	Assessed during the Comprehensive Examination through questions based on the classic Turing paper assigned for the course. These 10 marks are separate from the 30 marks for the Comprehensive Examination.
Tutorial Participation	10	Assessed during designated tutorials. Students will be informed whenever a tutorial is being used for assessment.

Important notes.

- Mid-semester exam: There is no mid-semester examination.
- Labs: Labs do not carry a separate evaluation component. They are conducted with the help of the TAs to connect the theory discussed in class with modern AI/LLM-based tools and methods.
- GitHub repo: Students should maintain a GitHub repository containing their worksheet/lab solutions and submit the repository at the end of the course.
- Tutorial: During tutorials, we will solve problems. Difficulty with material from a previous worksheet or lecture may be discussed during the following tutorial.
- Project: Project work is to be carried out in groups of no more than 4 students.
- NC (Not Cleared): Students are expected to participate sufficiently in the prescribed evaluation components to enable assessment of their performance. An NC report may be awarded in circumstances specified in the BITS Pilani Academic Regulations. Such cases will be handled in accordance with the applicable Institute and AUGSD procedures.

5. Bonus Marks Scheme

In addition to the evaluation above, students may opt into a bonus-marks scheme based on reading AI-related books. Bonus marks will be awarded only if the lecturers are satisfied that the student has genuinely read the book.

Non-fiction: any or all of

- Yuval Noah Harari, *Nexus* - 3 marks
- Judea Pearl, *The Book of Why* - 3 marks
- Arvind Narayanan and Sayash Kapoor, *AI Snake Oil* - 2 marks

Fiction: at most two of

- Isaac Asimov, *Bicentennial Man* - 1 mark
- Arthur C. Clarke, *2001: A Space Odyssey* - 1 mark
- Philip K. Dick, *Do Androids Dream of Electric Sheep?* - 1 mark
- Peter Watts, *Blindsight* - 1 mark
- D. F. Jones, *Colossus* - 1 mark

If the lecturers determine that a student has not read a selected book and has instead relied on a summary from another source (human or machine), the student will receive a penalty equal to that book's maximum bonus value (that is, $-X$ for a book worth at most X marks).

6. Text and Other Useful Books

Primary text

- S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Prentice Hall, 2021.

Other useful books

- Chip Huyen, *AI Engineering*, O'Reilly, 2024.
- Sebastian Raschka, *Build a Large Language Model (From Scratch)*, Manning, 2024.
- R. E. Neapolitan and X. Jiang, *Artificial Intelligence with an Introduction to Machine Learning*, 2nd ed., Chapman and Hall, 2018.

Instructor(s)

CS F407 - Artificial Intelligence